

SOURABH BHOSALE

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EDUCATION

Dr. Vishwanath Karad MIT World Peace University (MIT-WPU), Pune, India
B.Tech in Electronics and Communication Engineering

Aug 2022 – Jul 2026

WORK EXPERIENCE

RRC, IIT Hyderabad — Research Intern (Advised by [Prof. Dr. Spandan Roy](#) & [Soham Patil](#))

Aug 2025 – Present

- Implemented **DextraH-RGB** and **Maniwhere** policies sim-to-real for a 7-DoF xArm7 + 16-DoF LEAP Hand.
- Mapped oracle states to a 5D PCA space using **Dagger distillation** and **Geometric Fabrics** for reactive, collision-free control.
- Optimized URDF/USD assets and tuned **ADR** in **Isaac Lab** to mitigate low-torque hardware limits.
- Solved camera-shift degradation via a **view-invariant framework** fusing multi-view contrastive and 3D spatial objectives for uncalibrated monocular manipulation.
- NoContactNoWorries**: Developed a **Transformer-based pseudo-tactile sensing architecture** fusing **RGB-D, proprioception, and temporal motion cues** via pose-conditioned cross-attention to predict fingertip contacts without tactile sensors.
- Validated **sim-to-real contact prediction and closed-loop in-hand reorientation** on a physical **LEAP Hand**.

MIT Tech Team — Robocon Team of MIT WPU — Embedded Robotics Programmer

Sept 2022 – Jul 2026

- Leadership**: Electronics Head (2025), introduced agile practices such as Scrum and, as project manager for the robots, coordinated with mechanical, electronics, and software sub-teams and kept the build on schedule.
- Embedded Libraries**: Developed Libraries for STM32 and TI ARM Cortex-M4 hardware, building, wiring, and bridging high-level software to low-level actuation, integrating ODrive, VESC, and Cytron motor drivers over the CAN bus.
- Teaching**: Mentor for new members, imparting knowledge and learning experience to new beginners at MIT Tech Team.

PUBLICATIONS AND PROJECTS

NoContactNoWorries: Estimating Contact through Vision and Proprioception for In-Hand Dexterous Manipulation

- Accepted to IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2026*

DD Robocon (2023 Finalist, 2024 Semi-Finalist, 2025 Dunk Robot, 2026 Finalists)

- Led electronics architecture, embedded programming, mechanical prototyping and hardware debugging. Achieved AIR 2 (2023), AIR 3 (2024) and AIR 2 (2026) among 100+ national teams. Developed a jump-and-dunk capable robot for 2025 Robocon.

Object Invariant Anthropomorphic Dexterous Grasping (Final Year Project)

Jan 2026 – July 2026

- Trained a Reinforcement Learning policy of LEAP Hand + xArm 7 which is capable of grasping objects of various geometries using Vision + Proprioception + Tactile inputs. Used Isaac Lab for training RL policy and sim evaluation on Isaac Sim.

Development of ROS Based Catapult Arm

Nov 2024 – Dec 2024

- Designed and implemented a ROS2-based catapult arm capable of detecting, tracking, and shooting in Robocon 2024. Integrated RGB camera for visual perception, object detection using YOLO v8 and motor control for aiming, with accurate real-time operation.

Swerve Drive System

Sept 2023 – May 2024

- Implemented a three-wheel swerve drive system with precise motion control, integrating motor controllers, MEMS IMU, odometry, and encoders to enhance performance. Achieved a considerable improvement in speed, accuracy and repeatability during autonomous navigation.

Single Battery System

Oct 2023 – Feb 2024

- Designed a single-battery system to power multiple isolated robot subsystems, optimizing energy distribution and minimizing losses. Included overcurrent and overvoltage protection for stable operation under varying loads.

SKILLS

Robotics & Simulation:

NVIDIA Isaac Lab/Sim, Sim-to-Real Transfer, Teacher-Student Distillation, PPO, Geometric Fabrics, ROS 2, Gazebo

Computer Vision & Deep Learning:

Reinforcement Learning, Representation Learning (ResNet, LSTMs), Multi-View Contrastive Learning

Embedded & PCB Designing:

STM32, TM4C, AVR MCUs; CAN, UART, SPI, I2C, PWM; BLDC Motors; Altium, EagleCAD

Programming Languages & Tools:

C, C++, Python, Linux, Git, PyTorch